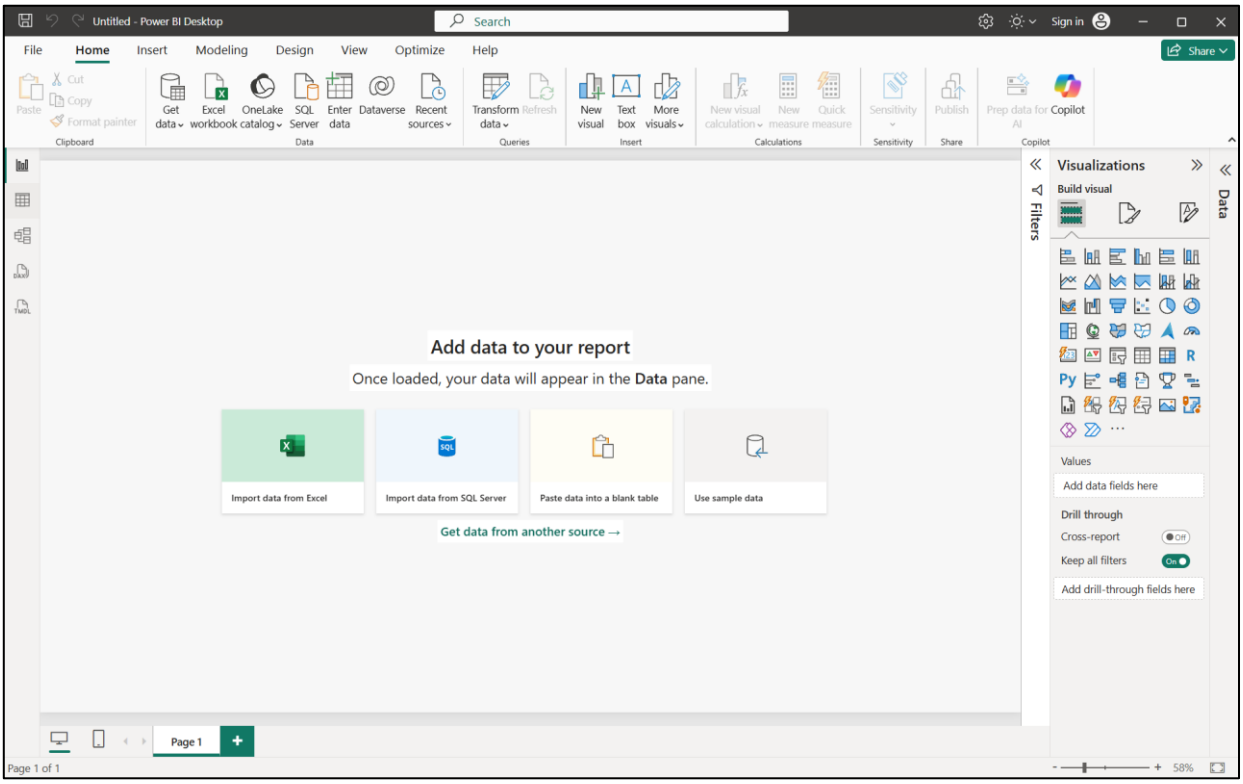
Name: Balleras, Vhon Xander R.	**Date:** November 2-6, 2026
Course: Data Analytics NCIII	**Trainer:** Christian G. Igay

Power BI Activities

Balleras, Vhon Xander Raz - Car Sales 5000 (Power BI)

car_id	sale_date	customer_name	gender	annual_income	dealer_name	company	model	engine	transmission	color	price	dealer_no	body_style	phone_no	dealer_region
1	1/29/2026	Karen Jones	Male	65000	Premier Motors	Honda	Accord	Electric	Automatic	Green	20447.27	D002	Sedan	-5688	North Carolina
2	7/28/2025	Sandra Miller	Female	86000	Capital Auto Sales	Honda	CR-V	Electric	Manual	Black	76428.17	D010	SUV	-9292	Michigan
3	8/8/2024	Susan Rodriguez	Male	53000	Summit Car Sales	Chevrolet	Silverado	I4	Manual	Silver	64486.9	D005	Sedan	-9928	Ohio
4	3/21/2024	Robert Brown	Female	177000	Premier Motors	Hyundai	Kona	I4	Automatic	Blue	77526.26	D002	Sedan	-6343	Connecticut
5	4/9/2025	Charles Jackson	Male	124000	Prime Vehicle Hub	Nissan	Sentra	V6	CVT	Gray	72040.16	D007	Coupe	-3486	Rhode Island
6	6/16/2024	Mary Moore	Female	172000	Cityline Automotive	Ford	Mustang	V8	Automatic	Blue	81278.8	D003	Hatchback	-6450	California
7	8/5/2024	Charles Martinez	Female	147000	Evergreen Autos	BMW	5 Series	V8	Automatic	Blue	75364.15	D004	Pickup	-8942	Pennsylvania
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...



TASKS:

1. Total Sales:

- YTD Total Sales
- PYTD Total Sales
- Sales Difference
- Sales Diff Color
- YoY Sales Growth
- MTD Total Sales
- MTD KPI

2. Average Sales:

- YTD Avg Price
- PYTD Avg Price
- Avg Price Difference
- Avg Price Color
- YoY Avg Price Growth
- MTD Avg Price
- MTD Avg Price KPI

3. Car Sold:

- YTD Car Sold
- PYTD Car Sold
- Car Sold Difference
- Car Sold Color
- YoY Car Sold Growth
- MTD Car Sold
- MTD Car Sold KPI

Power BI Learnings

Working on this Power BI activity using the Sales, Customers, and Products tables was a valuable hands-on learning experience for me. It helped me understand how data visualization and business intelligence work beyond simply creating charts and graphs. Through this activity, I learned how to transform raw data into meaningful information that can help people understand business performance and make better decisions.

One of the biggest things I learned during this activity was how to use DAX (Data Analysis Expressions) to create measures and calculate important sales metrics. At first, working with calculations such as YTD Total Sales, PYTD Total Sales, Sales Difference, and YoY Sales Growth felt challenging. However, as I continued working on the measures, I gradually understood how DAX allows us to compare sales performance across different periods. I also learned the importance of understanding the difference between current and previous year values when analyzing business trends.

Another important part of the activity was calculating Average Sales and Car Sold metrics. Creating measures for YTD Average Price, PYTD Average Price, Average Price Difference, and YoY Average Price Growth helped me understand how to analyze more than just total revenue. I also worked with MTD Total Sales, MTD Average Price, MTD Car Sold, and their corresponding KPIs. These measures showed me how useful time-based calculations are in monitoring performance and identifying whether a business is meeting its targets.

I also gained a better understanding of how conditional formatting and color indicators can improve data visualization. Using measures such as Sales Diff Color, Avg Price Color, and Car Sold Color helped me see how visual elements can communicate information more effectively. Instead of simply displaying numbers, Power BI can highlight positive and negative changes, making it easier for users to quickly understand the results. Learning how to create KPIs also helped me appreciate the importance of presenting data in a clear and meaningful way.

Overall, completing this Power BI activity strengthened my understanding of data analytics and business intelligence. It taught me that analyzing data is not only about calculating figures but also about presenting them in a way that is easy to understand and useful for decision-making. From creating DAX measures to analyzing year-to-date and month-to-date performance, I gained more confidence in using Power BI. This experience also encouraged me to continue improving my skills in data visualization, reporting, and analytics, which I can apply to future projects and real-world situations.

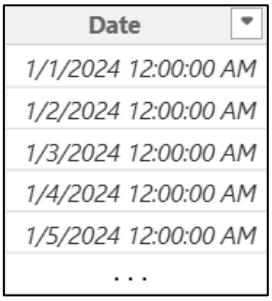
Vhon Xander R. Balleras
Trainee

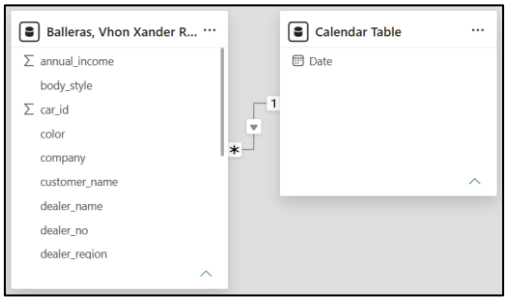
PREPARATION:

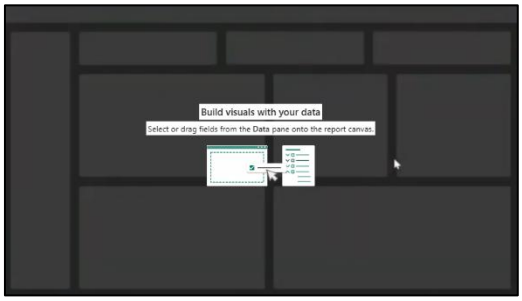
Creating Calendar Table, Relationship and Visualization.

```
1  Calendar Table = CALENDARAUTO()
```

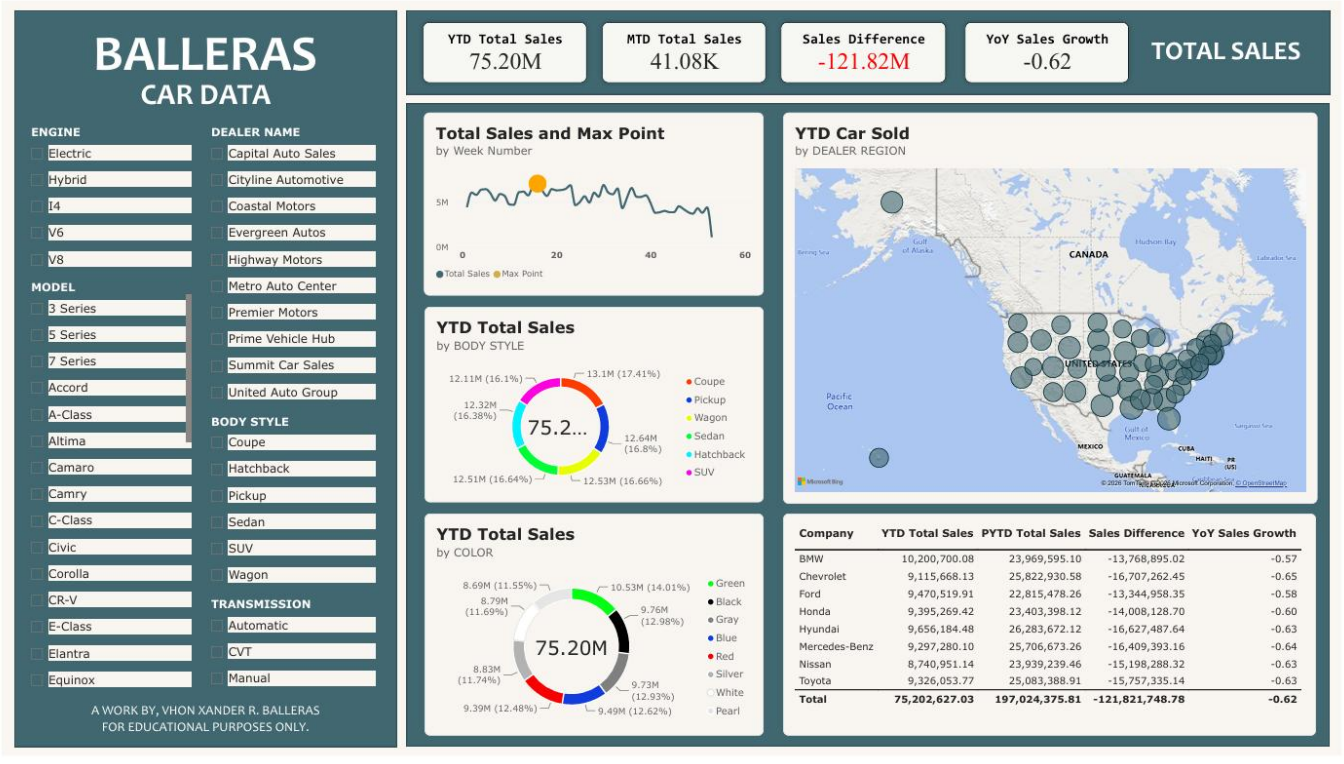
The formula creates a Calendar Table that automatically contains dates based on the dates found in the data. The CALENDARAUTO() function detects the date range in the dataset and generates a complete calendar, including all the dates within that range.







TASK 1: TOTAL SALES



TASK 1a: Total Sales - YTD Total Sales.

```
1  YTD Total Sales = TOTALYTD(SUM('Balleras, Vhon Xander Raz - Car Sales 5000 (Power BI)'[price]), 'Calendar Table'[Date])
```

The formula calculates the total sales from the beginning of the year up to the selected date. The SUM() function adds all the car prices, while TOTALYTD() calculates the accumulated sales for the year. The Calendar Table provides the dates used in the calculation.

TASK 1b: Total Sales - PYTD Total Sales.

```
PYTD Total Sales = CALCULATE(SUM('Balleras, Vhon Xander Raz - Car Sales 5000 (Power BI)'[price]), SAMEPERIODLASTYEAR('Calendar Table'[Date]))
```

The formula calculates the total sales for the same period in the previous year. The SUM() function adds all the car prices, while CALCULATE() changes the calculation based on the date range provided by SAMEPERIODLASTYEAR(). The Calendar Table helps Power BI identify the corresponding dates from the previous year.

TASK 1c: Total Sales - Average Difference.

```
Sales Difference = [YTD Total Sales] - [PYTD Total Sales]
```

The formula calculates the difference between the current year's total sales and the previous year's total sales for the same period. It subtracts PYTD Total Sales from YTD Total Sales to determine whether sales have increased or decreased. If the result is positive, sales have increased; if negative, sales have decreased.

TASK 1d: Total Sales –Sales Difference Color.

```
Sales Difference Color = IF([Sales Difference]> 0, "Green", "Red")
```

The formula determines the color based on the Sales Difference. The IF() function checks whether the Sales Difference is greater than zero. If it is, the formula returns "Green," indicating that sales have increased. Otherwise, it returns "Red," indicating that sales have decreased or remained the same.

TASK 1e: Total Sales – YoY Sales Growth.

```
YoY Sales Growth = [Sales Difference] / [PYTD Total Sales]
```

The formula calculates the year-over-year (YoY) sales growth by dividing the Sales Difference by the Previous Year-to-Date (PYTD) Total Sales. It shows how much sales have increased or decreased compared to the same period last year, expressed as a percentage. A positive result means sales have grown, while a negative result means sales have declined.

TASK 1f: Total Sales – MTD Total Sales.

```
MTD Total Sales = TOTALMTD(SUM('Balleras, Vhon Xander Raz - Car Sales 5000 (Power BI)'[price]), 'Calendar Table'[Date].[Date])
```

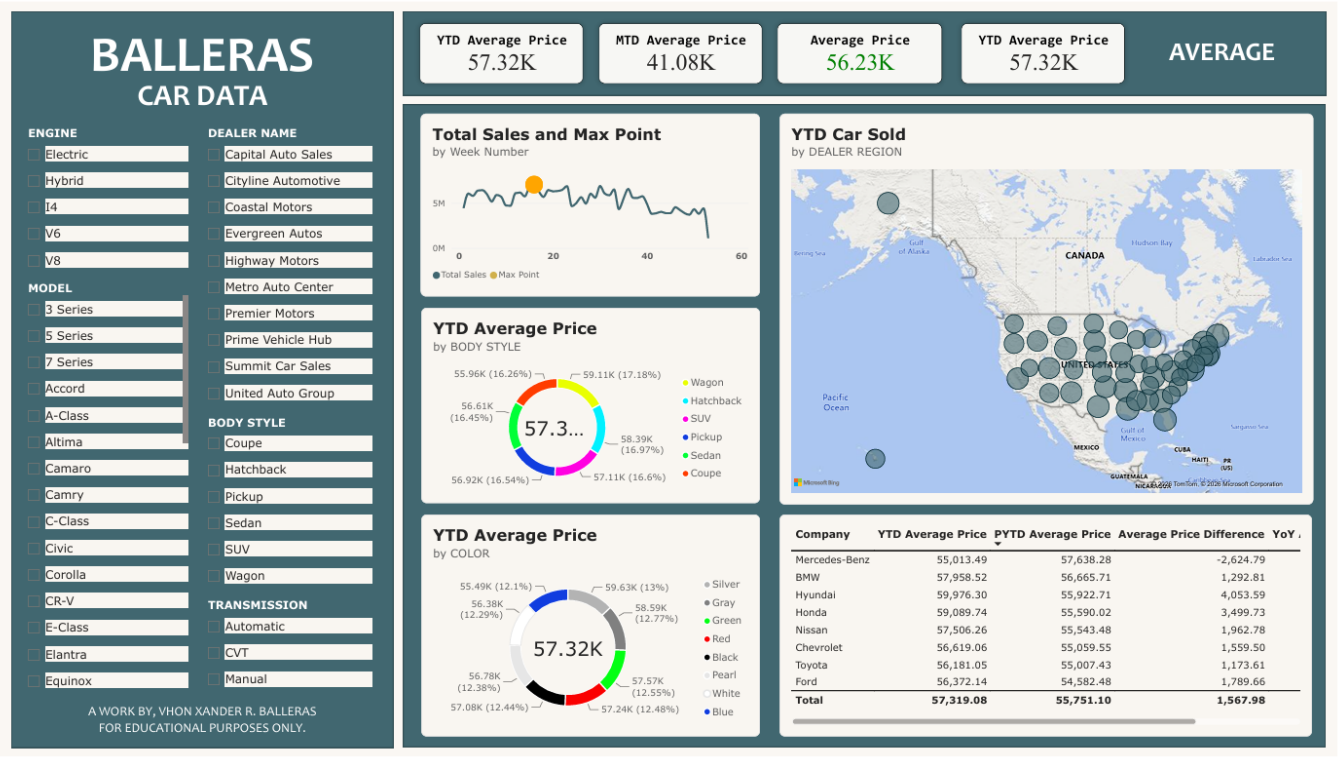
The formula calculates the total sales from the beginning of the current month up to the selected date. The SUM() function adds all the car prices, while TOTALMTD() calculates the accumulated sales for the month using the dates from the Calendar Table.

TASK 1g: Total Sales – MTD KPI.

```
MTD KPI = CONCATENATE("MTD Total Sales:", FORMAT([MTD Total Sales] / 1000000, "0.00M"))
```

The formula creates a text label that displays the Month-to-Date (MTD) Total Sales in millions. The CONCATENATE() function combines the text "MTD Total Sales:" with the calculated sales value, while FORMAT() changes the number into a readable format with two decimal places and an "M" to represent millions

TASK 2: AVERAGE PRICE



PREPARATION: Average Price.

```
1  Average Price = AVERAGE('Balleras, Vhon Xander Raz - Car Sales 5000 (Power BI)'[price])
```

The formula calculates the average price of all the cars in the dataset. The AVERAGE() function adds up the values in the price column and divides the total by the number of car records.

TASK 2a: Average - YTD Avg Price.

```
1  YTD Average Price = TOTALYTD([Average Price], 'Calendar Table'[Date])
```

The formula calculates the average car price from the beginning of the year up to the selected date. The TOTALYTD() function applies the year-to-date calculation to the [Average Price] measure using the dates from the Calendar Table.

TASK 2b: Average - PYTD Avg Price.

```
1  PYTD Average Price = CALCULATE([Average Price], SAMEPERIODLASTYEAR('Calendar Table'[Date]))
```

The formula calculates the average car price for the same period in the previous year. The CALCULATE() function changes the calculation context, while SAMEPERIODLASTYEAR() uses the Calendar Table to identify the corresponding dates from last year.

TASK 2c: Average – Avg Price Difference.

```
1  Average Price Difference = [YTD Average Price] - [PYTD Average Price]
```

The formula calculates the difference between the current year's average car price and the previous year's average car price for the same period. It subtracts PYTD Average Price from YTD Average Price to determine whether the average price has increased or decreased. A positive result means the average price has increased, while a negative result means it has decreased.

TASK 2d: Average – Avg Price Color.

```
1  Average Price Difference Color = IF([Average Price Difference]>0, "Green", "Red")
```

The formula determines the color based on the Average Price Difference. The IF() function checks whether the difference is greater than zero. If it is, the formula returns "Green," indicating that the average car price has increased. Otherwise, it returns "Red," indicating that the average car price has decreased or remained the same.

TASK 2e: Average – YoY Avg Price Growth.

```
1  YoY Average Price Growth = [Average Price Difference] / [PYTD Average Price]
```

The formula calculates the year-over-year (YoY) average price growth by dividing the Average Price Difference by the Previous Year-to-Date (PYTD) Average Price. It shows the percentage increase or decrease in the average car price compared to the same period last year. A positive result means the average price has increased, while a negative result means it has decreased.

TASK 2f: Average – MTD Avg Price.

```
1  MTD Average Price = TOTALMTD([Average Price], 'Calendar Table'[Date])
```

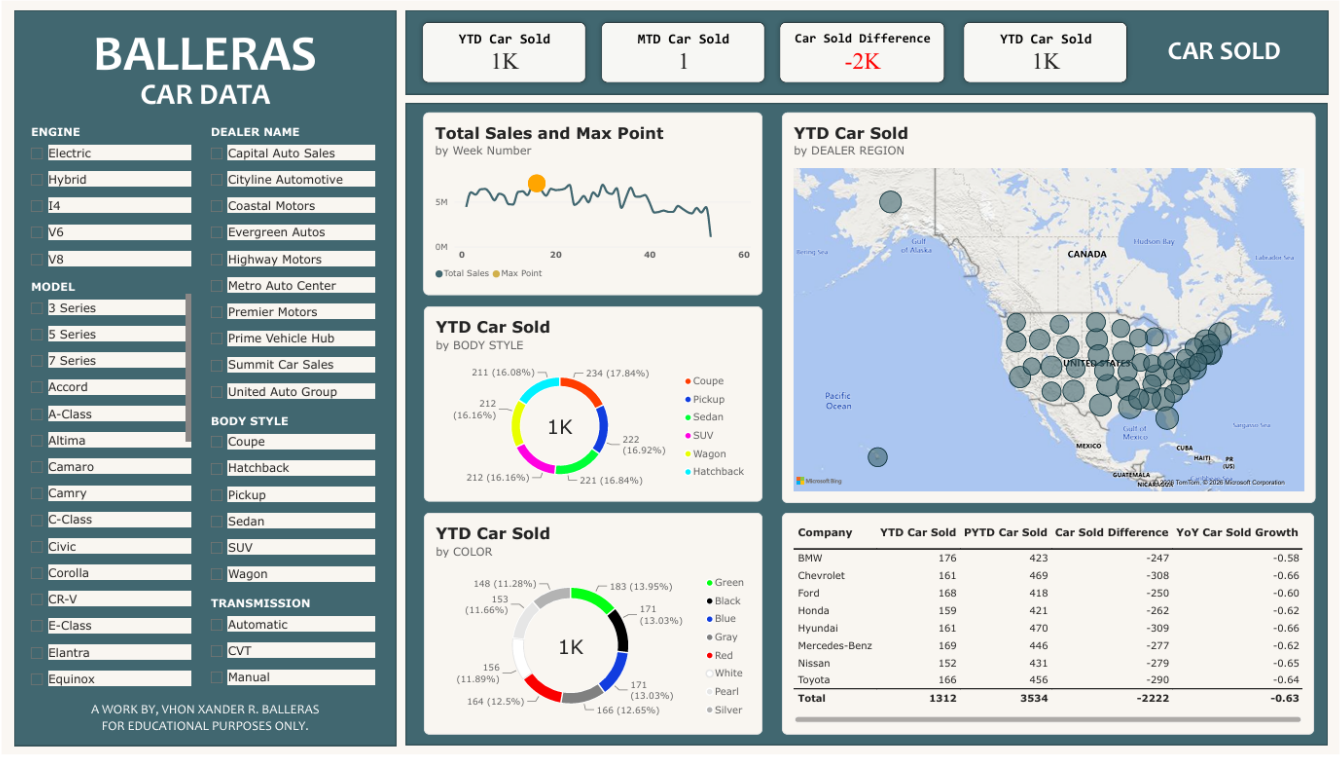
The formula calculates the average car price from the beginning of the current month up to the selected date. The [Average Price] measure calculates the average price of the cars, while TOTALMTD() applies the month-to-date date range using the Calendar Table.

TASK 2g: Average – MTD Avg Price KPI.

```
1  MTD Average Price KPI = CONCATENATE("MTD Average Price:", FORMAT([MTD Average Price]/1000, "0.00K"))
```

The formula creates a text label that displays the Month-to-Date (MTD) Average Price in thousands. The CONCATENATE() function combines the text "MTD Average Price:" with the calculated value, while FORMAT() converts the number into a readable format with two decimal places and a "K" to represent thousands.

TASK 3: CAR SOLD



Company	YTD Car Sold	PYTD Car Sold	Car Sold Difference	YoY Car Sold Growth
BMW	176	423	-247	-0.58
Chevrolet	161	469	-308	-0.66
Ford	168	418	-250	-0.60
Honda	159	421	-262	-0.62
Hyundai	161	470	-309	-0.66
Mercedes-Benz	169	446	-277	-0.62
Nissan	152	431	-279	-0.65
Toyota	166	456	-290	-0.64
Total	**1312**	**3534**	**-2222**	**-0.63**

TASK 3a: Car Sold - YTD Car Sold.

```
YTD Car Sold = TOTALYTD(COUNT('Balleras, Vhon Xander Raz - Car Sales 5000 (Power BI)'[car_id]), 'Calendar Table'[Date])
```

The formula calculates the total number of cars sold from the beginning of the year up to the selected date. The COUNT() function counts the car IDs in the dataset, while TOTALYTD() calculates the accumulated number of cars sold for the year using the dates from the Calendar Table.

TASK 3b: Car Sold - PYTD Car Sold.

```
PYTD Car Sold = CALCULATE(COUNT('Balleras, Vhon Xander Raz - Car Sales 5000 (Power BI)'[car_id]), SAMEPERIODLASTYEAR('Calendar Table'[Date]))
```

The formula calculates the total number of cars sold during the same period in the previous year. The COUNT() function counts the car IDs, while CALCULATE() changes the calculation based on the dates provided by SAMEPERIODLASTYEAR(). The Calendar Table helps Power BI identify the corresponding dates from last year.

TASK 3c: Car Sold – Car Sold Difference.

```
Car Sold Difference = [YTD Car Sold] - [PYTD Car Sold]
```

The formula calculates the difference between the number of cars sold this year and the number of cars sold during the same period last year. It subtracts PYTD Car Sold from YTD Car Sold to determine whether the number of cars sold has increased or decreased. A positive result means more cars were sold this year, while a negative result means fewer cars were sold.

TASK 3d: Car Sold – Car Sold Color.

```
1 Car Sold Difference Color = IF([Car Sold Difference]>0, "Green", "Red")
```

The formula determines the color based on the Car Sold Difference. The IF() function checks whether the difference is greater than zero. If it is, the formula returns "Green," indicating that more cars were sold this year. Otherwise, it returns "Red," indicating that fewer cars were sold or the number remained the same.

TASK 3e: Car Sold – YoY Car Sold Growth.

```
1 YoY Car Sold Growth = [Car Sold Difference] / [PYTD Car Sold]
```

The formula calculates the year-over-year (YoY) growth in the number of cars sold by dividing the Car Sold Difference by the Previous Year-to-Date (PYTD) Car Sold. It shows the percentage increase or decrease in the number of cars sold compared to the same period last year. A positive result means more cars were sold, while a negative result means fewer cars were sold.

TASK 3f: Car Sold – MTD Car Sold.

```
1 MTD Car Sold = TOTALMTD(APPROXIMATEDISTINCTCOUNT('Balleras, Vhon Xander Raz - Car Sales 5000 (Power BI)'[car_id]), 'Calendar Table'[Date])
```

The formula calculates the total number of unique cars sold from the beginning of the current month up to the selected date. The APPROXIMATEDISTINCTCOUNT() function estimates the number of unique car IDs without counting duplicates, while TOTALMTD() calculates the accumulated count for the current month using the Calendar Table.

TASK 3g: Car Sold – MTD Car Sold KPI.

```
1 MTD Car Sold KPI = CONCATENATE("MTD Car Sold:", FORMAT([MTD Car Sold] / 1000, "0.00K"))
```

The formula creates a text label that displays the Month-to-Date (MTD) number of cars sold in thousands. The CONCATENATE() function combines the text "MTD Car Sold:" with the calculated value, while FORMAT() converts the number into a readable format with two decimal places and a "K" to represent thousands.